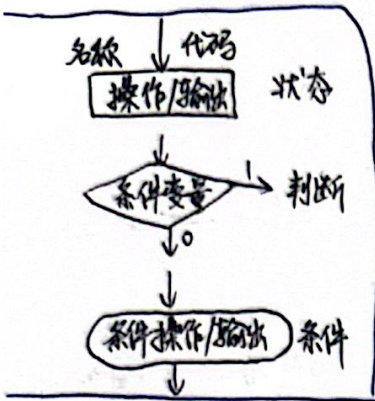
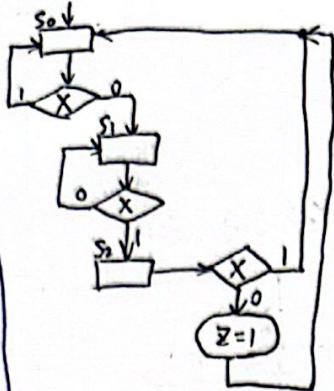
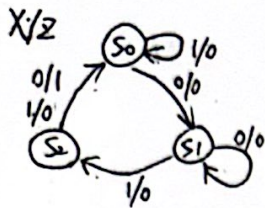


控制器

一. ASM图

1. 串行数据序列 X 输入 检测到 010 时 输出 Z

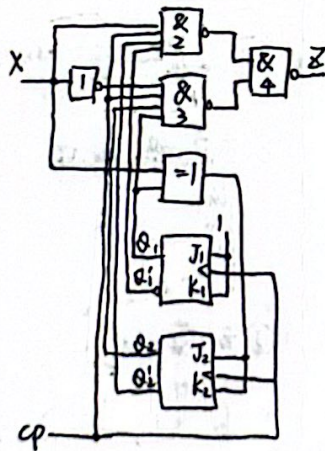
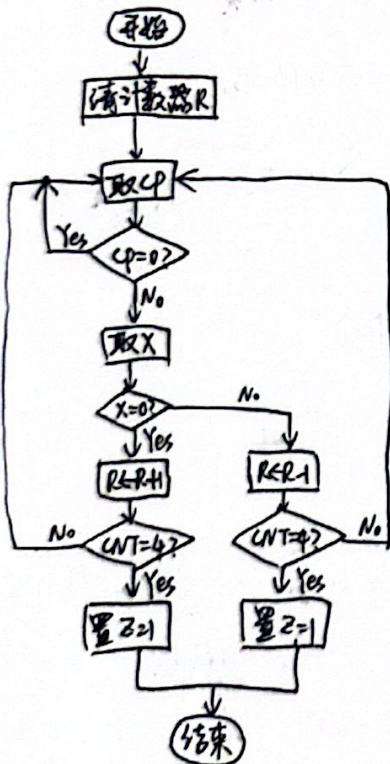
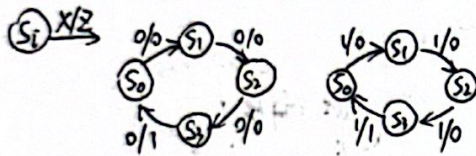
S₀ 起始 S₁ 收到 0 后 S₂ 收到 01 后 Mealy 型



2. 米里型同步模 4 可逆计数器

X=0 累加 4 个脉冲 输出进位 Z=1
X=1 累减 4 个脉冲 借位 Z=1

$X=0 \begin{cases} R \leftarrow R+1 & i=3 \text{ 时 } Z=1 (i=0 \sim 3) \\ R \leftarrow R-1 & i=0 \text{ 时 } Z=1 (i=0 \sim 3) \end{cases}$



二. 状态机设计

1. 给 ASM 图, 设计逻辑电路

① 状态分配

S₀~S₄ 000~110 0₂0₁0₀

② 激励表

0 ₂ 0 ₁ 0 ₀	00	01	11	10
S ₀	0	0	0	1
S ₁	1	X	X	X

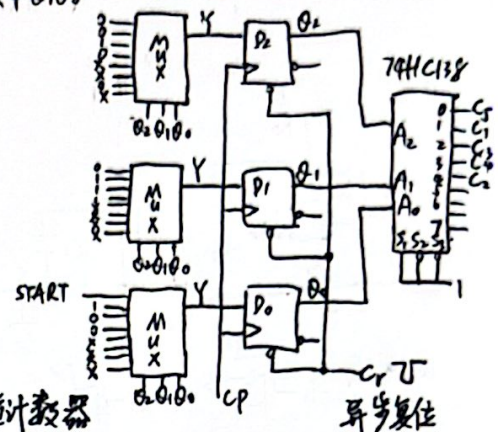
$$D_2 = \bar{0}_2 \bar{0}_1 \bar{0}_0$$

$$D_1 = 0_2 K + \bar{0}_2 0_1 + 0_0$$

$$D_0 = \bar{0}_2 \text{START} + 0_2 K + \bar{0}_2 0_0$$

0 ₂ 0 ₁ 0 ₀	00	01	11	10
D ₂	0	0	0	1
D ₁	1	X	X	X
D ₀	0	1	1	1
START	1	0	0	0
K	X	X	X	X

③ 电路图



④ 输出方程

$$C_5 = S_0 = \bar{0}_2 \bar{0}_1 \bar{0}_0$$

$$C_4 = S_3 = \bar{0}_2 0_1 0_0$$

$$C_3 = S_2 = \bar{0}_2 0_1 0_0$$

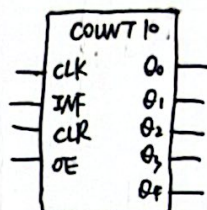
$$C_2 = S_4 = 0_2 0_1 0_0$$

$$C_1 = S_1 = \bar{0}_2 0_1 0_0$$

2. 异步清零

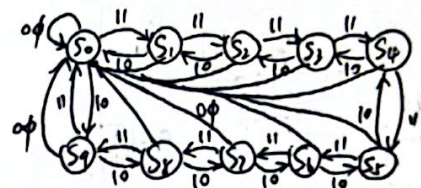
十进制加/减可逆计数器

① 确定任务



INF=1 加 0 减
OE=0 输出允许 Z
CLR=0 清零

② 状态转换图



③ 电路图

4 个 D 触发器 0₃0₂0₁0₀



CS 扫描全能王

3 亿人都在用的扫描 App

